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Printing apparatus for printing on cut sheets and roll sheets

Abstract

A printing apparatus includes a conveyance unit configured to be able to convey a roll sheet that is wound in a rolled shape and a cut sheet that is not wound in a rolled shape to a printing unit configured to print an image; a receiving unit detachably attached to the apparatus, the receiving unit being configured to receive the cut sheet onto which printing has been performed by the printing unit; a detection unit configured to detect whether the receiving unit is attached to the apparatus; and a holding unit configured to, in a case where the detection unit detects that the receiving unit is not attached to the apparatus, nip to hold the cut sheet onto which printing has been performed by the printing unit.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a printing apparatus.

Description of the Related Art

(2) Some printing apparatuses can handle various sheet sizes that comply with a predetermined standard such as A sizes (e.g., A0 size and A1 size), B sizes (e.g., B0 size and B1 size), and the like. Also, although some printing apparatuses comprise a placement unit (also expressed as receiving unit, tray, stacker, etc.) for placing sheets that are discharged (hereinafter, discharge sheets) from the main body of an apparatus, it is conceivable that in printing apparatuses that can handle various sizes, this placement unit can be attached or detached in relation to the main body of an apparatus (refer to Japanese Patent Laid-Open No. 2002-338125).

(3) In the aforementioned printing apparatuses, unintended printing may be performed in a state in which the placement unit is not attached to the main body of an apparatus. In such a case, the discharge sheet, by dropping from the main body of an apparatus, may receive unnecessary damage, which could cause a decrease in printing quality.

SUMMARY OF THE INVENTION

(4) The present invention is a printing apparatus that can handle various sizes and has an exemplary objective of improving printing quality in printing apparatuses in which a placement unit for placing discharge sheets can attach or detach in relation to the main body of an apparatus.

(5) One of the aspects of the present invention provides a printing apparatus, comprising a conveyance unit configured to be able to convey a roll sheet that is wound in a rolled shape and a cut sheet that is not wound in a rolled shape to a printing unit configured to print an image, a receiving unit detachably attached to the apparatus, the receiving unit being configured to receive the cut sheet onto which printing was performed by the printing unit, a detection unit configured to detect whether the receiving unit is attached to the apparatus, and a holding unit configured to, in a case where the detection unit detects that the receiving unit is not attached to the apparatus, nip to hold the cut sheet onto which printing was performed by the printing unit.

(6) Further features of the present invention will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic side view illustrating an example of a configuration of a printing apparatus according to an embodiment.

(2) FIG. 2 is a flowchart indicating an example of a form of printing by the printing apparatus.

(3) FIG. 3A is a flowchart indicating an example of a form of printing by the printing apparatus.

(4) FIG. 3B is a flowchart indicating an example of the form of the printing by the printing apparatus.

(5) FIG. 4 is a schematic side view illustrating an example of a form of holding a discharge sheet.

(6) FIG. 5 is a block diagram illustrating an example of the configuration of the printing apparatus.

DESCRIPTION OF THE EMBODIMENTS

(7) Hereinafter, embodiments will be described in detail with reference to the attached drawings.

Note, the following embodiments are not intended to limit the scope of the claimed invention.

Multiple features are described in the embodiments, but limitation is not made to an invention that requires all such features, and multiple such features may be combined as appropriate.

Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

(8) (Configuration of Printing Apparatus)

(9) FIG. 1 illustrates an example of a configuration of a printing apparatus **1** according to an embodiment, and FIG. 5 illustrates the block diagram thereof. The printing apparatus **1** comprises a printing unit **11**, an accommodation unit **12**, a conveyance unit **13**, a placement unit **14**, an attachment unit **15**, a detection unit **16**, a display/operation input unit **17**, a control unit **18**, and a discharge basket **19**.

(10) In the printing unit **11**, a printhead that can execute printing onto a sheet SH may be used. It is assumed that in the present embodiment, the printing unit **11** performs printing by an inkjet method, and on a side that faces the sheet SH of the printing unit **11**, a plurality of nozzles for discharging ink in relation to the sheet SH is provided. As another embodiment, another printing method may be employed in the printing unit **11**.

(11) Also, the printing unit **11**, in the present embodiment, is configured by a so-called serial head method (multipass method) and can scan in a sheet widthwise direction by a carriage. As another embodiment, the printing unit **11** may employ a so-called line head method (single pass method) in which a plurality of nozzles are provided across the entirety of a region that corresponds to the widthwise direction of the sheet. The accommodation unit **12** is configured to be able to respectively accommodate, as a target to be selected as a printing medium (hereinafter, printing target), a cut sheet in a size that complies with a predetermined standard and a roll sheet which is a long sheet wound in a rolled shape. In the present embodiment, the accommodation unit **12** includes a cut sheet stacking unit **121a** that is configured such that a plurality of cut sheets can be stacked, and in a position that is different from the stacking unit **121a**, a roll sheet mounting unit **121b** onto which a roll sheet is mounted so as to be rotatable. Also, the cut sheet stacking unit **121a** is disposed further on a downstream side in a conveyance direction d1 than the roll sheet mounting unit **121b** (refer to FIG. 1). Note that as an example of the above predetermined standard, a JIS standard (Japanese Industrial Standards) can be given.

(12) The conveyance unit **13** includes a plurality of conveyance rollers **131a** to **131d** for conveying the sheet SH from the accommodation unit **12**. A pair of rollers **131a** conveys, by rotating while sandwiching the cut sheet, from the cut sheet stacking unit **121a** to a downstream side one sheet at a time. A pair of rollers **131b** conveys, by rotating while sandwiching the roll sheet, from the roll sheet mounting unit **121b** to a downstream side one sheet at a time. A pair of rollers **131c** is disposed on the downstream side of the rollers **131a** and **131b** and on the upstream side of the printing unit **11**. The pair of rollers **131c** conveys, by rotating while sandwiching the cut sheet conveyed from the cut sheet stacking unit **121a** or the roll sheet conveyed from the roll sheet mounting unit **121b**, the cut sheet or the roll sheet toward the printing unit **11** as the sheet SH. Also, a pair of rollers **131d** conveys, by rotating while sandwiching the sheet SH onto which printing was performed by the printing unit **11**, the sheet SH toward a discharge port **20**. Although details will be described later, the pair of rollers **131d** may function, by sandwiching, in a state in which rotation has stopped, the sheet SH discharged from the discharge port **20** (hereinafter, may be expressed as discharge sheet SH), as a holding unit that holds the discharge sheet SH. When holding, a predetermined pressing force may be applied from one roller in relation to the other roller of the pair of rollers **131d**. This holding is also expressed as nipping and the like, and in this regard, the pair of rollers **131d** can also be represented as a nip unit, nip rollers, and the like. Moreover, although it is assumed that the rollers **131d** that perform conveyance (discharging) of the sheet SH also serve a function of a holding unit in the present embodiment, configuration may be such that the printing apparatus **1** comprises the holding unit which does not perform conveyance of the sheet SH and of which the object is only to hold.

(13) On the upstream side of the rollers **131d** and on the downstream side of the printing unit **11**, a detection unit (second detection unit) **132** that detects the existence or absence of the discharge sheet SH is provided. The detection unit **132**, in addition to detecting whether or not the sheet SH was appropriately discharged from the discharge port **20** and although details are described later, can also detect whether or not the sheet SH that is held by the pair of rollers **131d** was removed by

a user. Alternatively or additionally, a torque sensor or the like may be provided with the rollers **131d** themselves as the detection unit **132**.

(14) Here, in a case where the roll sheet is the printing target, the roll sheet is cut by a cutter unit (not shown) at a position in accordance with the length of printing that is based on a print job (print data), and then the cut sheet is discharged from the discharge port **20** by the rollers **131d**. The above cutter unit may be configured to be able to couple with the carriage that scans the printing unit **11** or may be incorporated in the carriage. Also, the cutter unit may be configured to move itself by a driving source different from that of the carriage.

(15) Note that in this specification, it is assumed that the downstream side indicates a side in the conveyance direction d1 (forward direction) in the path in which the sheet SH is conveyed by the conveyance unit **13** and that the upstream side indicates a side in the opposite direction from the conveyance direction d1 in the conveyance path.

(16) The placement unit **14** is provided to be able to attach or detach in relation to the main body of the printing apparatus **1**. In a case where the placement unit **14** is attached onto the main body of the printing apparatus **1**, the discharge sheet SH may be placed on the placement unit **14**. The placement unit **14** is also expressed as a receiving unit, a discharge tray, a discharge stacker, and the like or may be represented simply as a tray, a stacker, and the like. Also, the placement unit **14** may be expressed as an acceptance unit and the like as it accepts the discharge sheet SH from the main body of the printing apparatus **1**.

(17) The attachment unit **15** is configured so that the placement unit **14** can attach in relation to the main body of the printing apparatus **1** and for example, may be configured so that the placement unit **14** can freely be inserted or extracted and/or locked.

(18) The detection unit **16** is provided on the attachment unit **15** and is configured to be able to detect whether or not the placement unit **14** is attached onto the main body of the printing apparatus **1**. In the detection unit **16**, a sensor that can detect by an electrical connection an existence or absence of the placement unit **14** may be used or a press-type switch that is pressed by the placement unit **14** being attached may be used.

(19) The display/operation input unit **17**, in addition to being able to display predetermined information, is configured to be able to accept an operation input by the user, and in the present embodiment, it is assumed that a touch panel display is used in the display/operation input unit **17**. As another embodiment, the display unit may be provided separately from the operation input unit, and the operation input unit may be a button and the like that can be pressed by the user. Also, in the present embodiment, the display/operation input unit **17** functions as a notification unit for performing a predetermined notification to the user, and in the display/operation input unit **17**, an audio source for sounding an alerting sound and the like may be additionally provided.

(20) The control unit **18** functions as a system controller that drives and controls a system of the printing apparatus **1** as a whole. For example, the control unit **18**, based on a print job inputted into the printing apparatus **1**, by driving and controlling the printing unit **11** and the conveyance unit **13**, performs printing onto the sheet SH while conveying the sheet SH. Note that the print job may be received from an external apparatus (e.g., a personal computer of the user, etc.) or may be inputted into the display/operation input unit **17**.

(21) For example, the control unit **18**, by driving and controlling each electric motor, performs printing onto the sheet SH by the printing unit **11** while driving the conveyance unit **13** to convey the sheet SH. After the printing is completed, the control unit **18** further drives the conveyance unit **13** to have the sheet SH, onto which printing was performed, discharged from the discharge port **20**, and can start the next printing if necessary.

(22) The control unit **18** may be realized by a single unit or may be realized by electrically connecting two or more units to be able to communicate with each other. Also, the control unit **18** may be configured by a semiconductor apparatus such as an ASIC (application specific integrated circuit) and a PLD (programmable logic device) or may be configured by a CPU (central

processing unit) and a memory. In other words, the function of the control unit **18** can be realized by both hardware and software.

(23) Although details will be described later, the control unit **18**, based on a detection result from the detection unit **16**, can perform the predetermined notification to the display/operation input unit **17**.

(24) The discharge basket **19** is provided beneath the discharge port **20**. The discharge basket **19** is configured to be foldable and when in use, can be in a state in which it is opened (hereinafter, use state) by a user operation and when not in use, can be in a state in which it is folded (hereinafter, non-use state) by a user operation. In other words, the discharge basket **19** can transition between the use state and the non-use state. For example, the discharge sheet SH whose sheet length is relatively large is difficult to place on the placement unit **14** and so the discharge basket **19** in the use state can accept from the discharge port **20** such a discharge sheet SH. In the discharge basket **19**, a detection unit (third detection unit) **191** that detects whether the state is the use state or the non-use state is provided.

(25) In the present embodiment, to facilitate understanding, it is assumed that in a case where the cut sheet is used as the printing target, the discharge sheet SH is placed on the placement unit **14** and that in a case where the roll sheet is used as the printing target, the discharge sheet SH is discharged into the discharge basket **19**.

(26) (Form of Printing by Printing Apparatus)

(27) FIG. **2** is a flowchart indicating an example of a form of printing by the above described printing apparatus **1**. FIG. **3A** and FIG. **3B** indicate details with regards to a portion of the steps described in FIG. **2**. It is assumed that the present flowchart is started in accordance with the printing apparatus **1** entering a state in which printing can be executed and is ended in accordance with the printing apparatus **1** entering a state in which printing cannot be executed. An overview of the present flowchart is that in a case where a predetermined condition is met, the discharge sheet SH is held by the pair of rollers **131d**. Note that the present flowchart is mainly realized by the control unit **18**, and for example, determination in each step is performed by the control unit **18** functioning as the determination unit. In step **S100** (hereinafter, indicated simply as “**S100**”; the same applies to other steps), it is determined whether or not a print job was inputted into the printing apparatus **1**. Here, to facilitate understanding, it is assumed that the print job instructs printing related to a single sheet SH, in other words, the discharge sheet SH as a product is a single sheet. In a case where a print job was inputted, the processing proceeds to step **S110**, and otherwise, the processing returns to **S100**. In other words, the printing apparatus **1** is in a standby state until a print job is inputted.

(28) In **S110**, it is determined whether the printing target is a cut sheet or a roll sheet. Generally, the print job, in addition to printing information that indicates printing content itself, includes related information such as information that indicates whether to perform printing and in relation to which printing medium, and **S110** is performed based on that related information. In a case where the printing target is the cut sheet, the processing proceeds to **S120**, and in a case of the roll sheet, the processing proceeds to **S150**.

(29) In **S120**, the pair of rollers **131a** is driven and controlled to convey a single cut sheet from the cut sheet stacking unit **121a**, and moreover, the cut sheet is conveyed toward the printing unit **11** by the pair of rollers **131c**.

(30) In **S130**, printing based on a print job is performed by the printing unit **11** in relation to the cut sheet while the cut sheet is conveyed by the pair of rollers **131c**.

(31) In **S140**, an operation of discharging, as the discharge sheet SH from the discharge port **20** by the pair of rollers **131d**, the cut sheet onto which printing was performed by the printing unit **11** is started. Then, the processing returns to **S100**.

(32) In **S150**, the pair of rollers **131b** is driven and controlled to convey a roll sheet from the roll sheet mounting unit **121b**, and moreover, the roll sheet is conveyed toward the printing unit **11** by

the pair of rollers **131c**.

(33) In **S160**, printing based on a print job is performed by the printing unit **11** in relation to the roll sheet while the roll sheet is conveyed by the pair of rollers **131c**.

(34) In **S170**, the roll sheet onto which printing was performed by the printing unit **11**, after being conveyed to the downstream side by the pair of rollers **131d**, is cut by the cutter unit, and then the cut sheet is discharged as the discharge sheet SH from the discharge port **20** by the pair of rollers **131d**. Then, the processing returns to **S100**.

(35) FIG. 3A indicates details of **S140**. **S140** includes **S1410** to **S1460** and an overview thereof is that discharging of the discharge sheet SH is completed or the discharge sheet SH is held based on the existence or absence of the placement unit **14**.

(36) In **S1410**, based on a detection result of the detection unit **16**, it is determined whether or not the placement unit **14** is attached onto the main body of the printing apparatus **1**. In a case where the placement unit **14** is attached onto the main body of the printing apparatus **1**, the processing proceeds to **S1420**, and otherwise, the processing proceeds to **S1430**.

(37) In **S1420**, the pair of rollers **131d** is driven to discharge from the discharge port **20** the entirety of the discharge sheet SH and then the processing returns to **S100**. Because the placement unit **14** is attached onto the main body of the printing apparatus **1**, the discharge sheet SH is placed on the placement unit **14**.

(38) In **S1430**, the pair of rollers **131d** is driven to partially discharge from the discharge port **20** the discharge sheet SH so as to enter a state in which an upstream end of the discharge sheet SH remains within the main body (within the discharge port **20**) of the printing apparatus **1**. The discharge sheet SH may be partially discharged so as to be held at least by the pair of rollers **131d**, and regions other than the upstream end may remain within the main body of the printing apparatus **1**. In the present embodiment, partial discharging is performed so that at least the upstream end of the discharge sheet SH remains in a region that can be detected by the detection unit **132** (e.g., region that faces the detection unit **132**). By this, it can be detected that the discharge sheet SH is removed by the user.

(39) In **S1440**, the discharge sheet SH that was partially discharged from the discharge port **20** in **S1430** is held by the pair of rollers **131d**. This is realized by the pair of rollers **131d** sandwiching the discharge sheet SH in a state in which the rollers do not rotate. By this, the discharge sheet SH is held in a state in which the sheet is partially exposed from the main body of the printing apparatus **1** without dropping from the main body (or the discharge port **20**) of the printing apparatus **1** (refer to FIG. 4).

(40) In **S1450**, the user is notified that the discharge sheet SH is in a state in which the sheet is held. This notification may be performed by displaying predetermined information on the display/operation input unit **17** and may alternatively or additionally be performed by sounding an alerting sound and the like.

(41) In **S1460**, based on the detection result of the detection unit **132**, it is determined whether or not the discharge sheet SH in the above held state was removed. In a case where the discharge sheet SH was removed by the user, the processing returns to **S100** and otherwise, the processing returns to **S1440**. In other words, until the discharge sheet SH is removed, execution of printing related to the next sheet SH by the printing unit **11** and execution of conveyance of the sheet SH by the conveyance unit **13** are prevented, and the next printing is not started.

(42) FIG. 3B indicates details of **S170**. **S170** includes **S1710** to **S1760** and an overview thereof is that discharging of the discharge sheet SH is completed or the discharge sheet SH is held based on the state of the discharge basket **19**.

(43) In **S1710**, based on the detection result of the detection unit **191**, it is determined whether the discharge basket **19** is in the use state or the non-use state. In a case where the discharge basket **19** is in the use state, the processing proceeds to **S1720**, and otherwise, the processing proceeds to **S1730**.

(44) In **S1720**, similarly to **S1420**, the pair of rollers **131d** is driven to discharge from the discharge port **20** the entirety of the discharge sheet SH and then the processing returns to **S100**. Because the discharge basket **19** is in the use state, the discharge sheet SH is received by the discharge basket **19**.

(45) In **S1730**, similarly to **S1430**, the pair of rollers **131d** is driven to partially discharge from the discharge port **20** the discharge sheet SH so as to enter a state in which at least the upstream end of the discharge sheet SH remains within the main body of the printing apparatus **1**.

(46) In **S1740**, similarly to **S1440**, the discharge sheet SH that was partially discharged from the discharge port **20** in **S1730** is held by the pair of rollers **131d**. By this, the discharge sheet SH is held in a state in which the sheet is partially exposed from the main body of the printing apparatus **1** without dropping from the main body of the printing apparatus **1**. Note that because the roll sheet is typically long and has a risk of contacting an installation surface (floor surface) of the printing apparatus **1** when discharged from the discharge port **20**, a length to discharge the discharge sheet SH may be set or changed in accordance with the height from the installation surface to the discharge port **20**.

(47) In **S1750**, similarly to **S1450**, by displaying predetermined information on the display/operation input unit **17**, the user is notified that the discharge sheet SH is in a state in which the sheet is held.

(48) In **S1760**, similarly to **S1460**, based on the detection result of the detection unit **132**, it is determined whether or not the discharge sheet SH in the above held state was removed. In a case where the discharge sheet SH was removed by the user, the processing returns to **S100** and otherwise, the processing returns to **S1740**. In other words, until the discharge sheet SH is removed, the next printing is not started.

SUMMARY

(49) By virtue of such a form of printing, because the discharge sheet SH, in a case where it may drop from the main body (or the discharge port **20**) of the printing apparatus **1**, is held by the pair of rollers **131d** serving as the holding unit, it becomes possible to prevent unnecessary damage to the discharge sheet SH due to the drop. Accordingly, by virtue of the present embodiment, it becomes possible to improve quality of printing by the printing apparatus **1**.

First Embodiment

(50) Although in the embodiment, to facilitate understanding, the print job instructs printing related to a single sheet SH, the same control may be performed with regards to a case where printing related to a plurality of sheets SH is instructed.

(51) For example, a case where printing in relation to N sheets of sheets SH, where N is an integer greater than or equal to two, is performed is considered. Here, in a case where K-th sheet SH, where K is an integer greater than or equal to 1 and less than N, is discharged in a state in which the placement unit **14** is not attached onto the main body of the printing apparatus **1**, the sheet SH is held by the pair of rollers **131d** (refer to **S1440** to **S1460**). In the meantime, execution of printing related to (K+1)-th sheet by the printing unit **11** and conveyance of the sheet by the conveyance unit **13** are prevented; in other words, the remaining printing is interrupted.

(52) Meanwhile, in the meantime, in a case where the above held K-th sheet is removed by the user, printing related to (K+1)-th sheet by the printing unit **11** and conveyance of the sheet by the conveyance unit **13** are executed; in other words, the remaining printing is resumed.

Second Embodiment

(53) Although in the embodiment, to facilitate understanding, a form in which two types, the cut sheet and the roll sheet, are the printing targets was exemplified, in a case where there is a plurality of cut sheet sizes and/or in a case where there is a plurality of roll sheet widths, the same control may be performed.

(54) For example, assuming that in a case of the sheet SH that is less than or equal to a predetermined size (width and/or length is less than or equal to a predetermined dimension), the

sheet SH is placed on the placement unit **14**, and otherwise, the sheet SH is discharged into the discharge basket **19**, changes may be added to the flowchart in FIG. **3A** and FIG. **3B**. In other words, even in a case of the roll sheet, so long as the size can be received by the placement unit **14**, control to proceed to **S1410** in FIG. **3A** may be performed. Similarly, even in a case of the cut sheet, if the size is difficult to receive by the placement unit **14**, control to proceed to **S1710** in FIG. **3B** may be performed. Alternatively, assuming that a plurality of types of placement units **14** are provided and that one that supports the sheet is attached onto the main body of the printing apparatus **1**, changes may be made to the above flowchart.

(55) These can be realized, for example, in **S1410**, by determining whether or not the size of the sheet SH to be the printing target is a size supported by the placement unit **14**. In other words, in a case where the size of the sheet SH to be the printing target is a size supported by the placement unit **14** and the placement unit **14** is not attached onto the main body of the printing apparatus **1**, the discharge sheet SH may be held by the pair of rollers **131d** (refer to **S1440** to **S1460**).

Third Embodiment

(56) In the aforementioned embodiment, the form in which in a case where the discharge sheet SH in a state in which it is held by the pair of rollers **131d** is removed by the user, the processing returns to **S100** (refer to **S1460** and **S1760**) and the next printing can be started was exemplified. Although removal of the discharge sheet SH may be performed manually by the user, this may cause creases, wrinkles, tears, and the like to the discharge sheet SH. Accordingly, removal of the discharge sheet SH may be performed by another method.

(57) For example, the user may use the display/operation input unit **17** to instruct to discharge from the main body of the printing apparatus **1** the above held discharge sheet SH. In accordance with this, the control unit **18** rotates the pair of rollers **131d**, and by this the above holding is disengaged and then the discharge sheet SH is discharged from the main body of the printing apparatus **1**. By this, the user can accept the discharge sheet SH.

(58) As another example, the user, in accordance with notification in **S1450**, can attach onto the main body of the printing apparatus **1** the placement unit **14**. Because the detection unit **16** can detect that the placement unit **14** is attached onto the main body of the printing apparatus **1**, the control unit **18** can rotate the pair of rollers **131d** in accordance with the detection by the detection unit **16** and have the discharge sheet SH discharged from the main body of the printing apparatus **1**.

(59) By virtue of the above form of the removal of the discharge sheet SH, it becomes possible to prevent unnecessary damage to the discharge sheet SH and therefore, it becomes possible to further improve quality of printing by the printing apparatus **1**.

Fourth Embodiment

(60) Although as previously described, sheets SH of various sizes may be used as the printing target, it is conceivable that their materials also vary. Accordingly, it is conceivable that something (e.g., tracing paper) configured by a material that is relatively difficult to place onto the placement unit **14** may be the printing target. In such a case, irrespective of the existence or absence of the placement unit **14**, holding of the discharge sheet SH by the pair of rollers **131d** may be performed. This can be realized by specifying the material of the sheet SH to be the printing target.

(61) For example, in a case where related information included in a print job indicates information that indicates a material of a printing medium, the above holding by the pair of rollers **131d** may be performed based on the related information. Alternatively, the printing apparatus **1** may further comprise a specification unit that specifies a material of the sheet SH that is being conveyed by the conveyance unit **13** and the above holding may be performed based on a result of the specification. In the specification unit, a publicly known optical sensor may be used.

(62) (Program)

(63) The present invention can also be implemented by processing of supplying a program configured to implement at least one function of the above-described embodiments to a system or

an apparatus via a network or a storage medium, and reading out and executing the program by at least one processor in the computer of the system or the apparatus. The present invention may also be implemented by a circuit (for example, ASIC) that implements at least one function.

(64) (Others)

(65) In the above description, the printing apparatus **1** that uses an inkjet printing method has been described as an example, but the printing method is not limited to the above-described forms. Further, the printing apparatus **1** may be a single-function printer having only a printing function, or a multifunction printer having a plurality of functions such as a printing function, a fax function, and a scanner function. Additionally, for example, a manufacturing apparatus for manufacturing color filters, electronic devices, optical devices, microstructures, or the like by a predetermined printing method may be applicable.

(66) Also, the term “print” in this specification is to be broadly interpreted. Accordingly, an aspect of the term “print” is regardless of whether or not the object to be formed on the printing medium is meaningful information such as characters, and graphics, and whether or not the object is visualized so that a human can visually perceive it.

(67) Additionally, the term “printing medium” is to be broadly interpreted, similarly to the term “print” described above. Accordingly, the concept of the term “printing medium” may include any member that can receive ink, such as cloth, plastic film, metal plates, glass, ceramics, resin, wood, and leather, in addition to paper to be commonly used.

(68) Furthermore, the term “ink” is to be broadly interpreted, similarly to the term “print” described above. Accordingly, the concept of “ink” can include, in addition to a liquid that forms an image, a figure, a pattern, or the like by being applied onto a print medium, additional liquids that can be used for processing a print medium, processing ink (for example, coagulation or insolubilization of colorants in ink applied onto a print medium), or the like.

(69) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(70) This application claims the benefit of Japanese Patent Application No. 2020-039214, filed on Mar. 6, 2020, which is hereby incorporated by reference herein in its entirety.

Claims

1. A printing apparatus comprising: a stacking unit onto which a plurality of cut sheets that are not wound in a rolled shape are stacked; a mounting unit onto which a roll sheet that is wound in a rolled shape is mounted; a first pair of rollers configured to convey the cut sheet, as a print medium, stacked on the stacking unit, or a sheet, as a print medium, pulled out of the roll sheet mounted on the mounting unit in a conveyance direction to a printing unit configured to print an image by ejecting a liquid; a second pair of rollers, provided downstream of the printing unit in the conveyance direction, configured to convey the print medium onto which printing by the printing unit has been completed; and a front wall provided with a discharge port through which the print medium conveyed by the second pair of rollers is discharged, wherein the printing apparatus is operable in a first state in which a portion for receiving the print medium discharged from the discharge port is provided on the outside of the front wall, and a second state in which the portion is not provided, and wherein, in the second state, the sheet, pulled out of the roll sheet, discharged from the discharge port is held by the second pair of rollers.

2. The printing apparatus according to claim 1, wherein the first state is a state in which the portion, provided outside on the outside of the front wall, receives the cut sheet discharged from the discharge port.

3. The printing apparatus according to claim 1, wherein the stacking unit is provided above the

mounting unit.

4. The printing apparatus according to claim 1, further comprising an operation input unit configured to accept an operation input by a user.

5. The printing apparatus according to claim 4, wherein the operation input unit includes a touch panel display.

6. The printing apparatus according to claim 1, further comprising a cutter configured to cut the sheet pulled out of the roll sheet.

7. The printing apparatus according to claim 1, further comprising a carriage configured to carry the printing unit.

8. The printing apparatus according to claim 1, wherein the printing unit is a print head that discharges ink.

9. The printing apparatus according to claim 1, further comprising a scanner function.
